

Physical Science Part 1

1. Which refers to the recorded observations and measurements in an explanation?

- A. Conclusion
- B. Data
- C. Graph
- D. Variables

2. How is hypothesis tested?

- A. By gathering enough information about the problem
- B. By making inferences
- C. By conducting experiment
- D. By formulating conclusion

3. Which summarizes a natural occurrence that may be constantly observed given the same conditions which may be expressed in words or in mathematical form?

- A. Conclusion
- B. Hypothesis
- C. Law
- D. Theory

4. Ruben prepares a solution by putting a pinch of orange crystals to water in a beaker, then by consistent stirring, the crystal dissolve. Which of the statements above are likely to be observed by Ruben?

- I. The water becomes orange.
- II. The color of the solution becomes darker.
- III. Crystals disappear from the bottom of the beaker.
- IV. The beaker and its contents become heavier as attiring continues.

- A. I and II only
- B. III and IV only
- C. I, II, and III only
- D. I, II, and IV only

5. Which of the following is NOT true about scientific method?

- A. Conclusion is the last step
- B. Scientific method may be applied in every life.
- C. Scientific method is a systematic way of solving problem.
- D. Scientific method is done by following a step-by-step procedure in an exact sequence.

6. Which apparatus will you use to measure the mass of an irregular solid correctly?

- A. Empty beaker
- B. Erlenmeyer flash
- C. Graduated cylinder with water
- D. Vernier caliper

7. To graph the distance covered and velocity of a car, which is the dependent variable? Where should be it plotted on a line graph?

- A. Distance, x axis
- B. Distance, y-axis
- C. Velocity, x-axis
- D. Velocity, y-axis

8. A student found a piece of metal. What can the student to do quickly determine if the metal might contain iron?

- A. Heat the metal
- B. Weight the metal
- C. Place the metal in water
- D. Place the metal near the magnet

9. A student dipped a strip of aluminum and a strip of iron in water. The student then exposed the strips to air for one week. Which was most likely observed one week later?

- A. The iron strip formed rust.
- B. The aluminum strip formed rust.
- C. Both of the metal strips formed rust.
- D. Neither of the metal formed rust.

10.What human activity has added the most carbon in the atmosphere?

- A. Burning fossils fuels
- B. Mining fossil fuel
- C. Increasing soil erosion
- D. Cutting down the rain forests

11. The Table below lists measurements tasks included in laboratory activities conducted by students in a Grade 9 science class. Which matches a measurement task with the correct measurement tools?

Measurement Task	Tools
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I. Add 10g of salt to 200 ml of water and stir	Triple-beam balance, 25 ml capacity beaker marked in 1 ml intervals
II. Add 20 ml of rubbing alcohol to 50 ml of water	250 ml capacity Erlenmeyer flask marked in 50 ml intervals
III. Add 3 ml of cooking oil to 10g of sugar	5 ml capacity pipette marked in 1 ml intervals, pan balance with masses petri dish
IV. Add 25 g sugar to 15 ml of water and heat until the sugar dissolves	100 ml capacity graduated cylinder marked in 25 ml intervals, spring scale, mortar

- A. I
- B. II
- C. III
- D. IV

12. Aubrey wants to track her summative tests scores over the course of the first grading period. Which of the following will BEST allow her to analyze her progress?

- A. Bar graph
- B. Line graph
- C. Pie graph
- D. Scatter graph

13. What will most likely occur if global warming will not be addressed?

- A. Lower sea levels
- B. Melting of ice caps
- C. More pollution in water
- D. More reflected sunlight off the ice pack

14. How is the relationship described when both quantities are increasing but one quantity increases faster than the other.

- A. Direct proportion
- B. Direct square proportion
- C. Inverse proportion
- D. Inverse square proportion

15. In your attempt to determine which is denser oil or water, you mixed 25 ml of oil with 25ml of water. What helpful qualitative data can you derive from the set-up?

- A. There are 50 ml of liquid in the container.

- B. Oil did not mix with water.
- C. Oil is on top of the water.
- D. There are 25 ml of water and 25 ml of oil in the container.

16. Which of the following is a compound?

- A. Cemented wall
- B. Dry ice
- C. Gold
- D. Tincture of iodine

17. Which is most likely to change the blue litmus paper to red?

- A. Ammonia in water
- B. Detergent solution
- C. Lime water
- D. Vinegar

18. Which can be classified as homogeneous mixture?

- I. Buko pie
- II. Blood
- III. Virgin coconut oil
- IV. Mango juice

- A. III only
- B. I and II only
- C. I, III, and IV only
- D. II, III and IV only

19. Which of the following is a physical property of non-metals?

- A. Malleability and ductility
- B. Low tensile strength
- C. High density
- D. High tensile strength

20. The yellow color of a highlighter is a _____ property.

- A. chemical
- B. heterogeneous
- C. homogeneous
- D. physical

21. An odorless and colorless matter boils at 100 °C and melts at 0 °C What inference can be drawn from this observation?

- A. It is a metal.
- B. It is a mixture
- C. It is a non-metal.
- D. It is a pure substance.

22. You want to clean a tarnished silver. Which one will you use?

- A. Baking soda
- B. Salt solution
- C. Sugar solution
- D. Vinegar

23. What will you add to your food to make it last longer?

- A. Baking soda
- B. Salt solution
- C. Tomato
- D. Vinegar

24. Why is it not advisable to drink soft drinks on an empty stomach?

- A. It could lessen your appetite.
- B. It could make you feel full faster.
- C. It increases the acidity of your stomach.
- D. It contains stimulant which causes dizziness.

25. Almost the entire mass of an atom is concentrated in the _____?

- A. electron
- B. neutron
- C. nucleus
- D. proton

26. How many protons are present if an atom has a mass number of 23 and atomic number of 11?

- A. 11
- B. 12
- C. 23
- D. 44

27. Carbon-12 atom has _____.

- A. 6 electrons, 6 protons, 6 neutrons
- B. 6 electrons, 12 protons, 6 neutrons
- C. 12 electrons, 6 protons, 6 neutrons
- D. 16 electrons, 6 protons, 6 neutrons

28. What happens to the atomic number during a chemical reaction?

- A. Changes
- B. Remains the same
- C. Changes the alternately
- D. Changes and then is restored

29. Which of the following statements about isotopes of hydrogen is correct?

- A. They have different atomic numbers.
- C. They have the same number of neutrons .
- B. They have the same number of protons.
- D. They have different number of electrons.

30. Which of the following has 16 electrons?

- A. $^{27}_{13}\text{Al}^{+3}$
- B. $^{35}_{17}\text{Cl}^{-1}$
- C. $^{32}_{16}\text{S}$
- D. $^{32}_{16}\text{S}^{-2}$

31. How many protons, neutrons and electrons are there in an atom of $^{197}_{79}\text{Au}$?

- A. 79 protons, 79 electrons, and 118 neutrons
- B. 79 protons, 79 electrons, and 197 neutrons
- C. 197 protons, 79 electrons, and 79 neutrons
- D. 79 protons, 79 electrons, and 158 neutrons

32. The electrons configuration of an atom of element R in the ground state is $1s^2 2s^2 2p^6 3s^2 3p^6 4d^9$. What is the element described?

- A. $_{48}\text{Cd}$
- B. $_{35}\text{Br}$
- C. $_{31}\text{Ga}$
- D. $_{27}\text{Co}$

33. What is the shape of the molecule BeCl_2 ?

- A. Linear
- B. Angular
- C. Trigonal planar
- D. Tetrahedral

34. The generalized electron configuration of this group of element is $[\text{Noble gas}] ns^2ns$. Identify its group and family.

- A. Group VII and Family A
- B. Group V and Family A
- C. Group III and Family B
- D. Group I and Family B

35. What is the position of $_{12}\text{Mg}$ in the periodic table of elements?

- A. Group I-A, Period 2
- B. Group III-A, Period 2
- C. Group II-A, Period 3
- D. Group II-B, Period 4

36. What happens when a metal react with a nonmetal ? Usually_____.

- A. a simple covalent bond is formed
- B. a coordinate covalent bond is formed
- C. positive and negative ions are formed
- D. there is sharing of electrons in the outer shells of two elements

37. What bond exists in a solid that does not conduct electricity but conducts electricity when dissolved in water?

- A. Coordinate covalent bond
- B. Covalent bond
- C. Ionic bond
- D. Metallic bond

38. Which of the following statements about a triple bond is CORRECT?

- A. Compounds with triple bonds are not reactive.
- B. An example of a molecule with a triple bond is C_2H_6 .
- C. Triple bond is weaker than single and double bonds.

D. Three pairs of electrons can hold two atoms together stronger than two pairs.

39. Which chemical bond will most likely exist if element A has an electron configuration of $1s^2 2s^2 2p^3$ and element B has $1s^2 2s^2 2p^5$?

- A. Coordinate covalent
- B. Ionic
- C. Metallic
- D. Polar covalent

40. X, Y, and Z represent elements of atomic numbers 9, 11, and 16. What type of bond will be performed between Y and Z?

- A. Ionic bond
- B. Metallic bond
- C. Nonpolar covalent bond
- D. Polar covalent bond

Answer Keys:

1. B
2. C
3. C
4. C
5. D
6. C
7. B
8. D
9. A
10. A
11. C
12. A
13. B
14. B
15. C
16. B
17. D
18. D
19. B
20. D
21. D
22. A
23. D
24. C
25. C
26. A
27. A
28. B
29. B
30. C
31. A
32. C
33. A
34. A
35. C
36. C
37. C
38. D
39. D
40. A